

The Sepsis Six: Does Public Data Show Whether It Worked?

Bootstrap CUSUM — ONS Sepsis Mortality England & Wales 2001–2023 — Executive Summary
stepchangeanalysis.com | May 2026 | N=23, Loops=5,000, Conf=90%

Four Key Findings

What the Sepsis Six is

Six bedside actions within one hour: oxygen, blood cultures, IV antibiotics, IV fluids, serum lactate, urine output monitoring. Developed 2005, Dr Ron Daniels. 96% of NHS trusts, 37 countries.

Bootstrap CUSUM result

Both public mortality series show one stage, no change point at 90% confidence across 22 years (N=23). No structural change detectable in either clinical mortality measure.

The smoking gun — the ratio

Bootstrap CUSUM on the ratio (mentioned anywhere / underlying cause) finds a structural change point at 2013, 95.5% confidence, -19.7%. The measurement system changed. The clinical outcome measure did not.

The PDSA failure

The NHS answered the second Langley question ("how will we know?") with process measures (compliance rates) not outcome measures (mortality rate). The Study step was never completed at national level.

The Ratio — Stage Summary

1	2001–2013	10.75	Baseline	Baseline
2	2013–2023	8.63	-19.7%	95.5%

The Three Measure Types — Langley et al., The Improvement Guide

■ Outcome What ultimately matters	Age-standardised mortality rate; in-hospital mortality (deaths/admissions); quality of life at 90 days.	■ Not reliably available. ONS series compromised by coding changes. HES available but CUSUM not applied. QoL not collected.
■■ Process Whether the change is being done	Bundle compliance %; time-to-antibiotics; patients screened.	■■ Partially available 2018+. NHS reported this as proof of success — but compliance ≠ outcome.
■ Balancing Unintended consequences	Post-sepsis syndrome (40–60% of survivors); antibiotic resistance; fluid overload.	■ Not systematically tracked nationally. Post-sepsis syndrome never reported alongside compliance data.

The Question That Matters Most for Patients

This article examines mortality. Broad et al. (2025) examines survival. Between the two, a picture emerges neither paper alone provides: more patients are surviving — but we do not know what that survival means for their lives. A patient wants three things:

① Not to die	ONS mortality data (this article) — partially addressed, coding-limited.
② Leave hospital alive	HES data (Broad et al. 2025) — addressed. Survival rates have improved.
③ Return to life before	Neither paper addresses this. 40–60% of survivors develop post-sepsis syndrome. Survival ≠ recovery.

The key sentence: The 2025 paper shows more patients are surviving sepsis. Our article shows we cannot confirm whether the Sepsis Six is why. Neither paper measures what survival actually means for patients — and that is the most important question of all.

Bright Spots — Where Is It Working and Why?

The national null result conceals variation — some trusts almost certainly outperforming others. Positive deviance (Zeitlin, 1990; Sternin & Sternin, Vietnam 1991; Pascale, Sternin & Sternin, 2010; Heath & Heath, 2010) asks: where is it working and what are those places doing differently? Bootstrap CUSUM applied at trust level — identifying trusts with genuine structural change points in their in-hospital mortality rates — is the rigorous tool for that analysis. The improvement opportunity lies in propagating the bright spots, not just pushing the national average.

Key References

- Daniels R et al. — The Sepsis Six. Emergency Medicine Journal, 2011;28(6):507–512.
- McPherson D et al. — 8.6% finding. BMJ Open, 2013.
- Broad J et al. — Rising rates of sepsis in England. Infection, 2025. doi:10.1007/s15010-025-02601-0. Guy's & St Thomas' NHS FT / King's College London / University College London.
- Langley G et al. — The Improvement Guide. 2nd ed. Jossey-Bass, 2009. Three measure types and Model for Improvement.
- Zeitlin M et al. — Positive Deviance in Child Nutrition. UN University Press, 1990.

- Pascale RT, Sternin J & M. — The Power of Positive Deviance. Harvard Business Press, 2010.
- Heath C & D. — Switch. Random House, 2010.
- Deming W.E. — Out of the Crisis. MIT Press, 1982.
- Joiner B.L. — Fourth Generation Management, Ch.8 p.128. McGraw-Hill, 1994.
- NICE NG51 — Sepsis recognition, diagnosis and early management, 2016 (updated 2019).
- NICE CG191 — Rehabilitation after critical illness. UK Sepsis Trust post-sepsis support: sepsistrust.org/patient-information
- Full article: stepchangeanalysis.com/sepsis-six-does-public-data-show-whether-it-worked.html